ResearchAgent

Research with a wallet.

An agent that answers from free sources first, then buys the one missing article straight from its writer, **over x402 on the XRP Ledger**, inside a budget you set. If the article breaks its promise, the money comes back.

- XRPL Testnet · no real money
- The writers and their articles are fictional

— What we built

One question, one budget, and only code can spend it

1

Ask

A question and a budget, S$0 to S$5.

2

Confirm the plan

A 5-second sheet. It confirms the plan, not a purchase.

3

Search writers

Each writer’s site returns abstracts and a signed fact menu, never premium text.

4

Decide

A decision model judges what is worth buying, weighted by trust.

5

Pay

Policy code pays the writer over x402 on XRPL, within S$1 per source.

6

Verify, refund, answer

Check every promise; refund a broken one; cite delivered passages.

Five hard gates, on every run

1. No premium bytes before a grant.
2. The budget is the only spending authority; only policy code can buy, never the LLM.
3. One charge per intent, across retries and restarts.
4. Every citation points at a delivered passage.
5. Anything simulated or substituted is labelled.

0 violations

Read-only gate audit (six rules): 10 runs, 728 observations, held-out evaluation, 3 Nov 2026. Enforced in CI by 687 tests in 83 files.

Progress since SingHacks

From a gated prototype to a product other agents can use

19–20 Sep
SingHacks
Prototype: a human approved every purchase.

10 Oct
AI Tinkerers SG
Live demo and decision-model benchmark; Testnet payments 7–9 Oct.

3 Nov
v0.1
MCP server and nightly scorecard.

3 Nov
v0.2
The app, keyboard-first, record mode.

3 Nov
v0.3
Signed fact menus from writers.

3 Nov
v0.4
Mac app: DMG, Keychain.

3 Nov
v0.5
The money, made clear (UX rebuild).

Merged, to tag
v0.6 +
Escrow, hire the agent, test USD, audit trail, agent allowances.

40
pull requests merged on 2–3 Nov, after the October demo

5
tagged releases, v0.1.0 → v0.5.0; the Mac DMG and demo clips are release assets

687
tests in 83 files on main (3 Nov), including every hard gate

35
builder-feedback items, each with evidence, a workaround and a suggested fix

gh release list · gh pr list --state merged (2–3 Nov SGT) · docs/SFF-PUSH.md changelog. Release dates in SGT; tags v0.6+ are TODO on the feature-day calendar.

— Reachability · 25%

Six ways in, for people and for other agents

Claude Code, or any MCP client

One `claude mcp add`. Four tools. The caps live in the environment; the calling model cannot raise them.

Testnet purchase from MCP: 0.09 XRP, 22.8 s, retry with the same key, no second charge.

Mac app

Apple Silicon DMG with Keychain-held seeds. Works offline for judges, no keys.

Installed launch to interactive: 2.07 s. DMG 130 MiB, unsigned until Developer ID.

SDK, TypeScript and Python

Ask, search, stream a run, read the wallet. There is no purchase method: only backend policy can buy.

Typed SDK in repo; npm publish is the owner's call.

Hire the agent over x402

Another agent pays ResearchAgent for an answer; it buys, pays cited writers and returns the change.

Five-hop money chain on Testnet (next slide).

Publish in one command

`npm run publish:site -- ./my-notes` turns a Markdown folder into a writer agents can find and pay.

Folder → verified paid read: 18.90 s, 0.01 Testnet XRP.

One wallet, many agents

Per-agent hashed keys with their own caps, pause and revoke, under one wallet limit.

Two MCP clients + SDK: 0.18 Testnet XRP, 0 violations.

> **Compliance readiness, stated plainly:** a hash-chained audit log written with every spend decision, CSV/JSONL receipt export, read-only `audit:verify --testnet`, a fail-closed payee denylist and data pruning. It is a Testnet prototype; no regulator or lawyer has reviewed it.

— Creativity · 20%

Agents as customers and as sellers: the citation list is a payment instruction

Hop on XRPL Testnet, 3 Nov 2026	Drops
Caller agent → ResearchAgent `8AD6FD7B…24296134`	220,000
ResearchAgent → NotFinancialTimes, article `FA422ABF…4F8726AE77`	90,000
Royalty to NotFinancialTimes, cited `AF86FDEB…3C86F7FE5`	2,500
Royalty to Open Records, a **free** writer, cited `48C600E3…E373E68BD95E`	2,500
Change back to the caller `CCDF6611…8ADEA4F39`	110,000
Service fee kept by ResearchAgent	15,000
Every drop accounted for	**220,000**

- **A new business model for writers.** Being cited earns a royalty, even when the article was free. A quarter of the service fee is split equally across cited writers.
- **The gates still hold.** The caller's payment is the budget; policy code makes every purchase; royalties are computed from the citation list, not chosen by a model.
- **Refused payouts refund the caller.** A denylisted writer's royalty was cancelled before signing and 2,500 drops went back, atomically.
- **Native XRPL only:** Payments with InvoiceID and memos binding the run. No contract to deploy.

Proof or refund, enforced by the ledger, not the writer’s goodwill

The buyer pays into a native XRPL Escrow with a crypto-condition. The secret is revealed only after the delivered article passes its signed proof.

Proof passed

EscrowCreate → proof verified → EscrowFinish 12.1 s

C82B337D…1099A25926 → 7367BA10…1D02BD4846

Proof failed

EscrowCreate → claim not found → CancelAfter 120 s → EscrowCancel 129.4 s

EAA86976…72444F60A72 → C007A296…E78408FA

> Full principal back with the writer **offline**. Limit we state: a hashlock is not fair exchange; a dishonest buyer could read and refuse, so writers opt in.

- **Persist before submit.** The signed blob is stored before it is sent; a restart reconciles the same hash and never signs twice.
- **Pending is a third state.** A missing transaction is not proof money didn’t move; expiry needs a validated ledger past LastLedgerSequence.
- **Invoice binding.** x402 invoice JSON → SHA-256 → XRPL InvoiceID, so a payment can only settle its own quote.
- **Exact delivery.** Partial payments are rejected before submit; `delivered_amount`, currency and issuer are checked after.
- **Prompt injection can’t spend.** Model inputs are capped and untrusted; purchases come only from policy code.
- **Tamper-evident audit.** A hash chain written in the same transaction as each reserve, sign, settle and refund.

Cheap enough per question, and it recovers from failure

US$0.0043

model cost per question

Live decision and coverage models, 92 held-out questions, 3 Nov

12 drops

network fee per payment

0.000012 Testnet XRP; escrow finish costs 423 drops

22.8 s

MCP question → paid, cited answer

Real Testnet settlement, fixture research

6.24 ms

queued payment p95, was 307.91 ms

Synthetic: 300 ms writer delay, same signer

15 s

reconciler cycle for in-flight payments

Durable journal; health and metrics; runbook

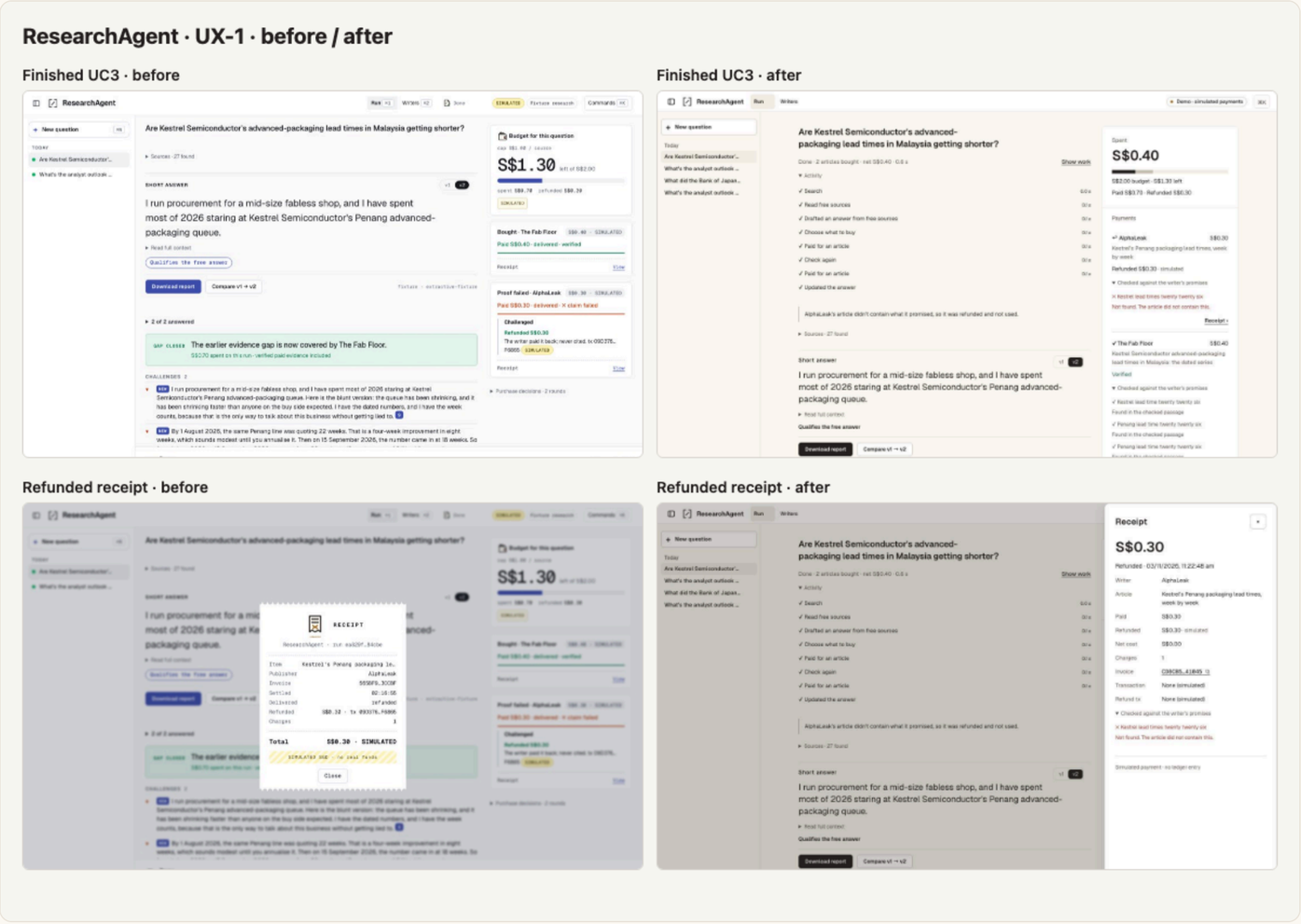
546 ms

cold start, demo API and publisher

Mac app installed: 2.07 s to interactive

> **Where it still falls short:** held-out wasted spend fell from 52.87% to **40.68%** (95% CI 11.2–68.4%) but misses our 25% target. Evaluation runs use live decision models on a synthetic corpus with simulated payments; we publish the miss with its interval.

The money is the hero, and it is plain



- **One environment indicator** in the header replaces label spam. “Simulated” stays on receipts, refunds and exports.
- **Pay → verify → refund** read as payment rows changing status in place. Each row opens its receipt and transaction hash.
- **A refund says why:** “AlphaLeak’s article didn’t contain what it promised. Refunded S$0.30.”
- **Motion only for money:** 30 keyframes down to 4, all under 240 ms.
- **Keyboard-first:** ⌘K, ⌘1 / ⌘2, and ⌘. to stop further buying.
- **Fast:** palette p95 30 ms, view switch p95 34 ms, layout shift 0.014.

35 items of builder feedback, each with evidence and a suggested fix

1 Ship a payment-recovery example, not just a payment

A restart between signing and delivery can sign twice. Durable intent → signed blob → validated ledger → delivery.

2 Timeouts need a third state: pending

A missing tx on one node is not proof money never moved. Mainnet can’t treat an RPC timeout as permission to pay again.

3 One invoice-binding fixture across x402 and XRPL

HTTP invoice JSON vs ledger InvoiceID hash: publish cross-language vectors, including malformed ones.

4 Make the refund trust boundary explicit

A ledger payment proves value moved, not that an article kept its promise. Escrow expiry is eligibility, not a refund.

5 Pricing units block mainnet

The RLUSD Testnet faucet returned HTTP 401; we used a labelled test-USD stand-in issuer rather than bypass it.

+ Mainnet readiness: a gap checklist, not a claim

Key custody (KMS or external signer), FX quotes, facilitator SLA, KYB and sanctions screening, history retention for audits.

docs/BUILDER-FEEDBACK.md: 20 findings from building + 15 September research items; dated, with evidence, severity and mainnet impact

— Check it yourself

Seven moments, seven validated Testnet transactions

What happened, 3 Nov 2026	Transaction
An MCP client buys one article within a S$2 cap	`D65FDA44…E32FC9F9`
A Markdown folder becomes a paid writer	`ED26E003…EB08F15FA5`
Escrow released after proof passed	`7367BA10…1D02BD4846`
Escrow cancelled after proof failed	`C007A296…E78408FA`
Another agent hires ResearchAgent	`8AD6FD7B…24296134`
Writer paid in an issued test USD	`E6AD8E14…05A711D01B`
Two agents, one wallet, separate caps	`281E67C5…F0F2B4C3`

Before mainnet

- Payer keys in a KMS or external signer
- Official RLUSD issuer and live FX quotes
- KYB and a real sanctions-screening provider
- Hosted durable storage and a full-history node
- Signed, notarised Mac app (Developer ID)

Repo · `github.com/Collaboration95/theFastandtheFungible`
Video · TODO: link after recording (target 11 Nov)